ORGNZ-02 LAB: Understanding Grail Buckets - Hands-on Exercises

> **Series:** ORGNZ | **Notebook:** 2 of 10 | **Type:** LAB | **Created:** February 2026 | **Last Updated:** 02/19/2026

Overview

This lab notebook contains 5 hands-on exercises extracted from **ORGNZ-02: Understanding Grail Buckets**. Complete the lecture notebook first, then work through these exercises to reinforce the concepts with real DQL queries against your Dynatrace environment.

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Prerequisites

Requirement	Details
Completed	ORGNZ-02: Understanding Grail Buckets (lecture notebook)
Dynatrace Environment	SaaS tenant with Grail enabled
Permissions	`logs.read`, `metrics.read`, `entities.read`, `spans.read`

Exercise 1: Bucket Administration Permissions

ORGNZ-02: Understanding Grail Buckets

> **Series:** ORGNZ | **Notebook:** 2 of 10 | **Created:** January 2026 | **Last Updated:** 01/28/2026

Buckets are logical storage containers in Dynatrace Grail where records are stored. Think of a bucket as a folder in a filesystem. Use buckets to group telemetry data that naturally belongs together, such as data from the same region, environment, or with the same sensitivity classification.

| Requirement | Details |
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1. What Are Buckets?
2. Supported Data Types (Tables)
3. Bucket Limits
4. Query Constraints
5. Default Buckets
6. Querying Bucket Information

7. Querying Data from Buckets
8. Bucket Characteristics
9. Bucket Naming Rules
10. When to Create Custom Buckets

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Dynatrace Environment	SaaS environment with Grail enabled
Permissions	`storage:buckets:read` for viewing, `storage:bucket-definitions:write` for creating
Knowledge	Completed ORGNZ-01

Permission	Purpose
`storage:bucket-definitions:read`	View bucket configurations
`storage:bucket-definitions:write`	Create and modify buckets
`storage:bucket-definitions:delete`	Remove buckets
`storage:bucket-definitions:truncate`	Truncate bucket data

By the end of this notebook, you will:
- Understand bucket fundamentals and supported data types
- Know bucket limits and query constraints
- Query bucket information using DQL
- Recognize when custom buckets are needed

Buckets organize data to meet performance, compliance, and retention requirements:

Function	Description
Data Grouping	Organize large datasets into meaningful segments
Streamlined Organization	Records that should be handled together are stored together
Query Performance	Reduced queried dataset improves query speed
Retention Management	Each bucket has its own retention policy
Access Control	Bucket-level IAM policies for team isolation
Cost Attribution	Enable data cost tracking by organizational unit

Each bucket is associated with exactly **one data type**:

Data Type	Default Bucket	Description
logs	`default_logs`	Log records from all sources
metrics	`default_metrics`	Metric data points
spans	`default_spans`	Distributed trace spans
events	`default_events`	Platform events
bizevents	`default_bizevents`	Business events
security_events	`default_security_events`	Security-related events

> **Important**: You cannot mix data types in the same bucket. A logs bucket only stores logs.

Limit	Value	Notes
Maximum buckets per environment	80	Default limit; can request increase
Typical capacity	Up to 5 TB/day per table	With default bucket limit

Metric	Value	Impact
Optimal ingest per bucket	~1 TB/day	Best query performance
Acceptable ingest	1-3 TB/day	Limited query window
Maximum ingest	3 TB/day	Performance degrades above this

Limit	Value
Minimum retention	1 day
Maximum retention	3,657 days (~10 years + 1 week)

Understanding query limits is critical for bucket planning:

Limit	Value	Impact
Maximum data scanned	500 GB	Limits queryable time window
Maximum records returned	1,000	Use aggregations for larger datasets
Maximum response payload	1 MB	Large result sets may be truncated

The 500 GB scan limit determines how far back you can query:

Daily Ingest	Queryable Window
500 GB/day	~24 hours
1 TB/day	~12 hours
3 TB/day	~4 hours
10 TB/day	~1 hour

> **Implication**: High-volume buckets may only allow querying recent data. Use aggregations, time filters, and strategic bucket partitioning to work within these limits.

Dynatrace provides default buckets with varying retention:

Default Bucket	Data Type	Default Retention	Notes
`default_logs`	logs	35 days	Standard log retention
`default_metrics`	metrics	15 months	Extended for trend analysis
`default_spans`	spans	10 days	Short-term APM data
`default_events`	events	35 days	Platform events
`default_bizevents`	bizevents	35 days	Business events
`default_security_events`	security_events	35 days	Security events

> **Note**: Query your environment's `dt.system.buckets` to verify current retention settings.

```
// List all buckets with retention and status
fetch dt.system.buckets
| fields name, display_name, dt.system.table, retention_days,
estimated_uncompressed_bytes
| sort dt.system.table asc, name asc
```

Exercise 2: Bucket Administration Permissions

```
// Query logs from default bucket (implicit)
fetch logs, from:-1h
| limit 10
```

Exercise 3: Bucket Administration Permissions

```
// Query logs from a specific custom bucket
fetch logs, bucket:{"default_logs"}
| limit 10
```

Exercise 4: Querying Data from Buckets

```
// Query from multiple buckets simultaneously
fetch logs, bucket:{"default_logs", "audit_logs", "security_logs"}
| limit 100
```

Exercise 5: Querying Data from Buckets

```
// Filter by bucket within query
fetch logs, from:-1h
| filter dt.system.bucket == "default_logs"
| filter loglevel == "ERROR"
| limit 50
```

Lab Summary

You have completed 5 hands-on exercises for **ORGNZ-02: Understanding Grail Buckets**.

Exercises Completed

- [ ] Exercise 1: Bucket Administration Permissions
- [ ] Exercise 2: Bucket Administration Permissions
- [ ] Exercise 3: Bucket Administration Permissions
- [ ] Exercise 4: Querying Data from Buckets
- [ ] Exercise 5: Querying Data from Buckets

Next Steps

Continue with **ORGNZ-03** for the next notebook.
